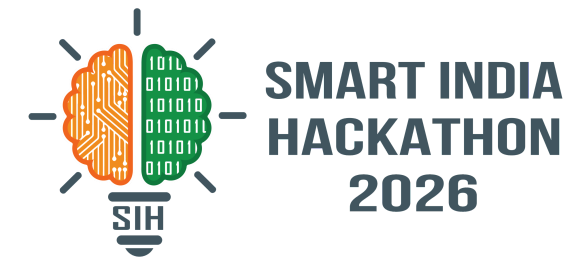


# SMART INDIA HACKATHON 2026



**Team :- Hudson Hackers**

**Problem Statement ID: SIH26067**

**Problem Statement Title:** Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations

**Theme:** Disaster Management

**PS Category:** Software

**Team Name:** Hudson Hackers.

**Project/Solution Name:** VARUNA -Visualization & Assimilation of Real-time Underwater Network Analytics

# VARUNA = Visualization & Assimilation of Real-time Underwater Network Analytics

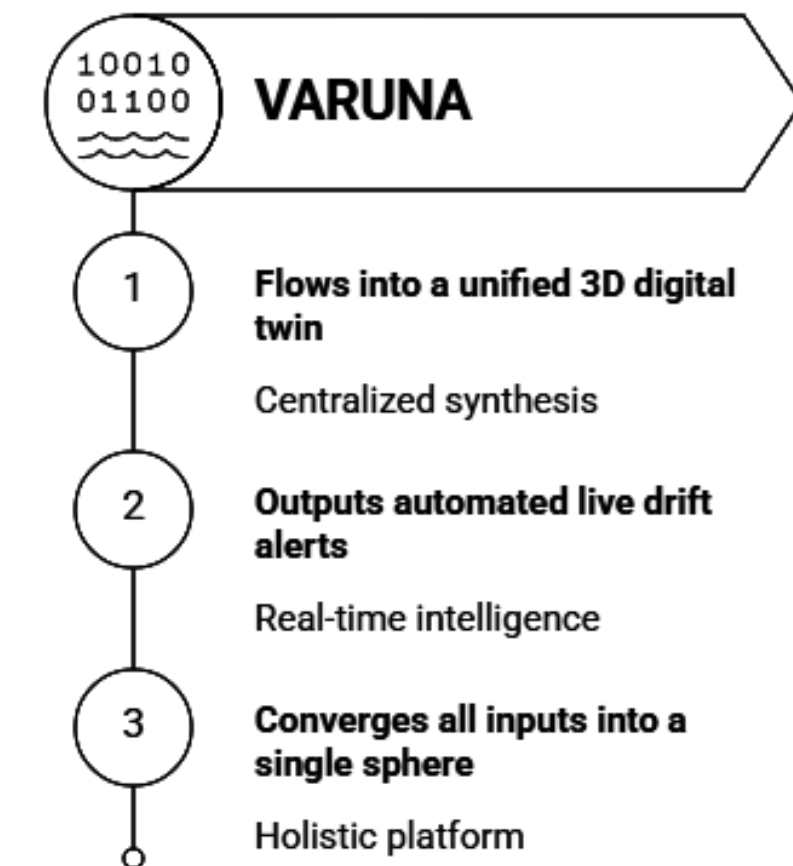
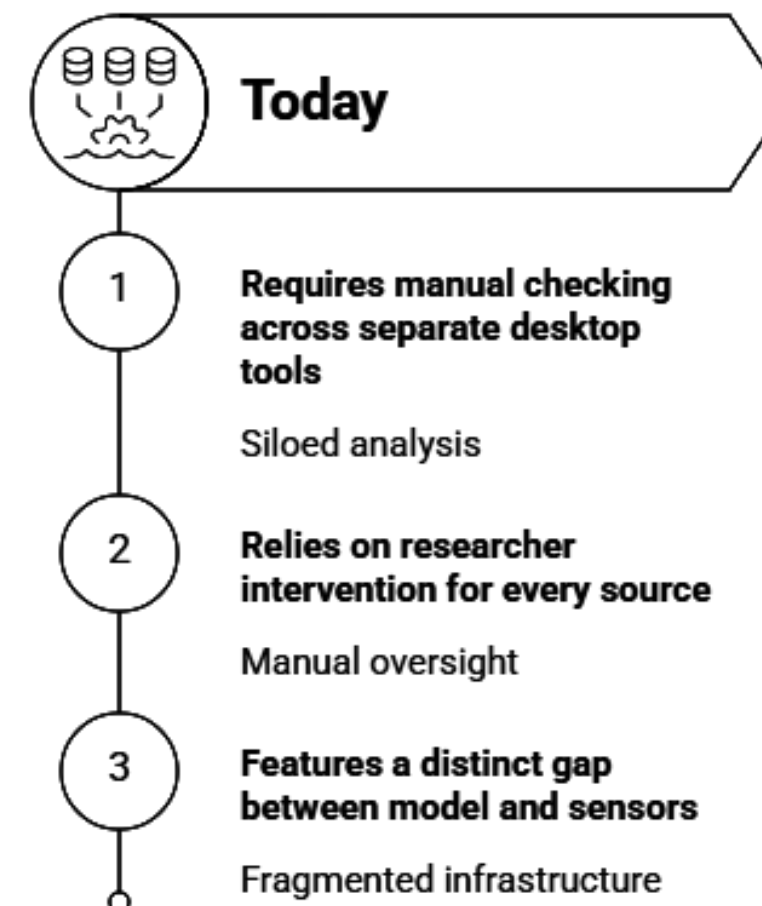
## OUR SOLUTION :-

A browser-based **3D platform** that fuses INCOIS's ROMS-based HOOFS model output with real **Argo, Glider, CTD, and buoy observations and actively flags** when the model and reality disagree, instead of leaving that comparison to manual, periodic desktop work.

## Innovation & Uniqueness

1. **Graph-based fusion**, not distance-based interpolation
2. **Live drift/anomaly alerts**, not manual validation
3. Built on named 2024-2026 frontier AI research & not a vague "AI-powered" label-
  - **Graph Neural Networks**
  - **Titans-style test-time memory**,
  - **ConvLSTm based nowcasting**
4. **Dual mode** - dense scientific view for researchers, simplified "safe/caution/avoid" view for fishermen and the public
5. First tool to **combine a true digital twin in 3D + live fusion + India-specific + dual-audience**.

## How should oceanographic data be processed and analyzed?





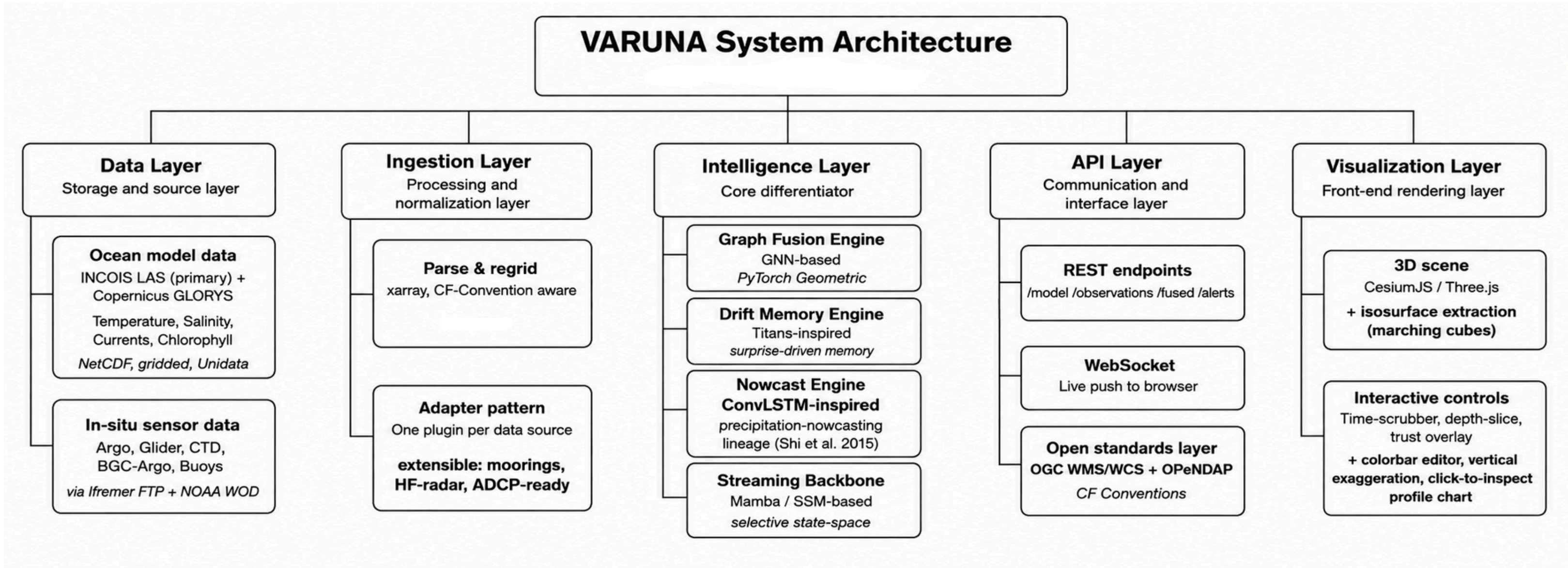


fig.System architecture

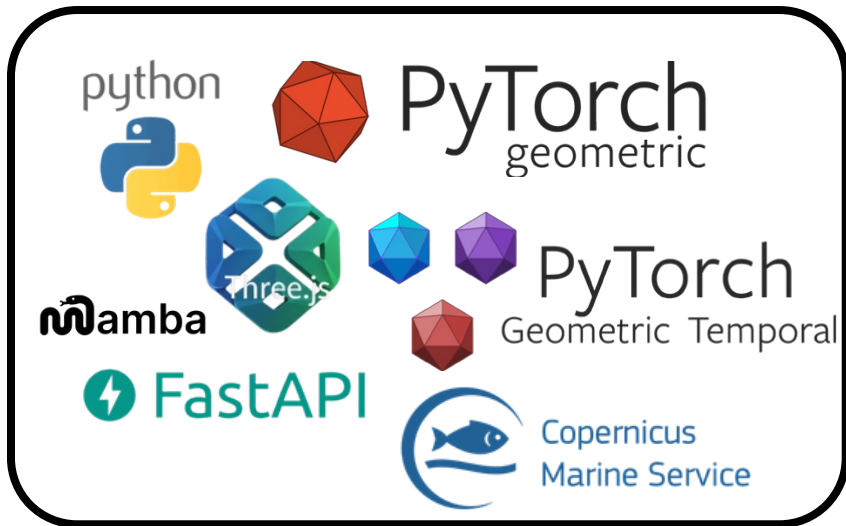
## Validation to be build on the above data

**Region:** Bay of Bengal

**Variables:** Sea Surface Temperature (SST), Currents, Wave Height, Salinity, Depth

**Data source:** Copernicus Marine (GLORYS) & NOAA HYCOM reanalysis INDOFOS-equivalent for the demo window

**Live validation against:** Argo, Glider, CTD, and buoy readings for the same region



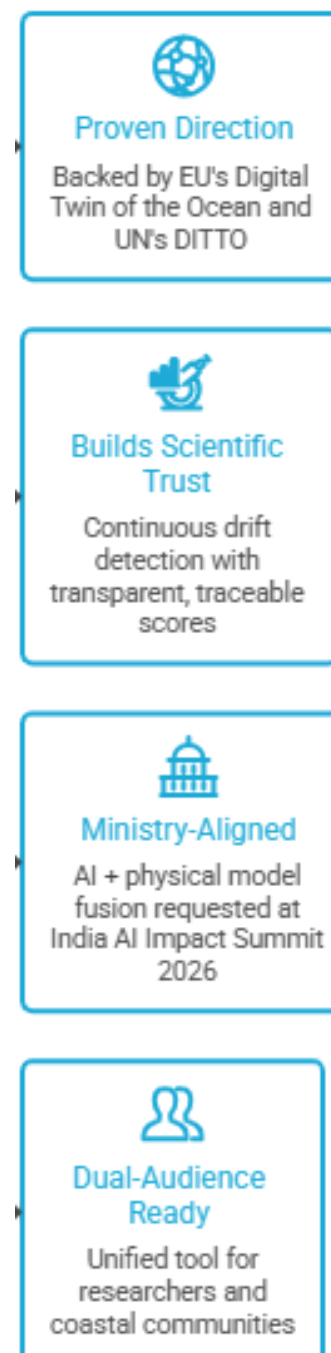
**Fuse → Detect → Validate → Serve → Show**



## FEASIBILITY ANALYSIS

- **Open, reusable stack:** Built on proven open-source tools and public data, no proprietary licenses needed
- **Scalable deployment:** Scoped to Bay of Bengal, extends to any INCOIS region with no architecture change
- **Cost-effective:** No new hardware runs entirely on data INCOIS already collects and publishes
- **Graceful fallback:** Every advanced feature has a tested lightweight backup, never breaks to zero

## VIABILITY



## RESEARCH IMPACT

- **Continuous validation:** Replaces periodic manual model-checking with an always-on comparison, freeing researcher time
- **Surfaces hidden signals:** Live drift alerts can catch marine heatwaves or current anomalies researchers might otherwise miss between manual checks
- **Reusable research tool:** The graph fusion approach isn't Bay-of-Bengal-specific - extends to any basin, any reanalysis product
- **Open API for science:** Fused, validated data exposed via REST/WebSocket means other researchers can build on top, not just view



### IMPACT

- 7,500+ km of Indian coastline - one platform built to watch all of it
- From once-a-season manual validation to live, continuous trust
- One alert reaches a fisherman and a research paper - at the same time
- Backed by India's ₹4,077 crore Deep Ocean Mission and \$1 trillion+ Blue Economy push
- Built for the people who need it most - fishermen, Navy, Coast Guard, and scientists, together

fig. Static geography to a living ocean twin

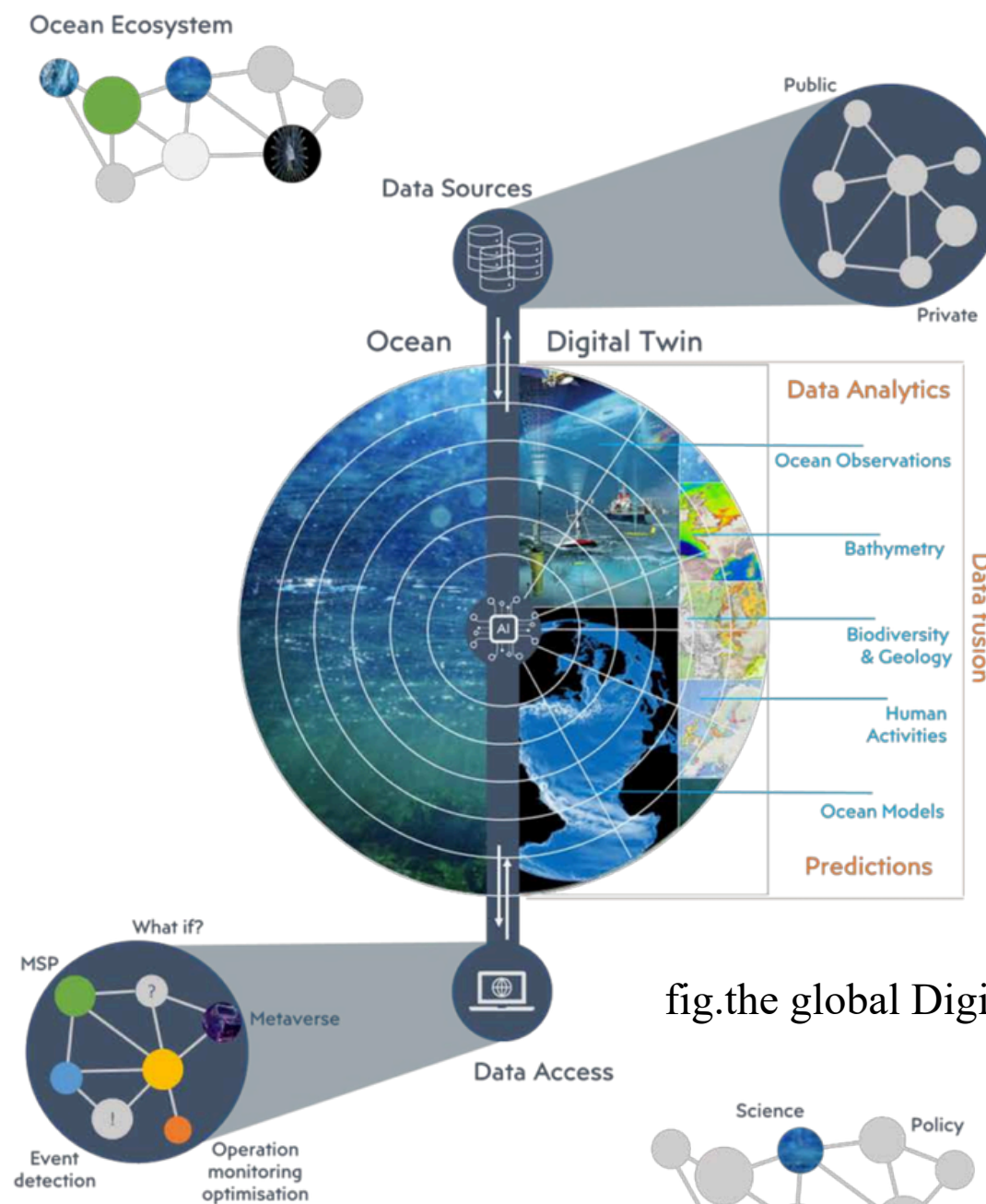
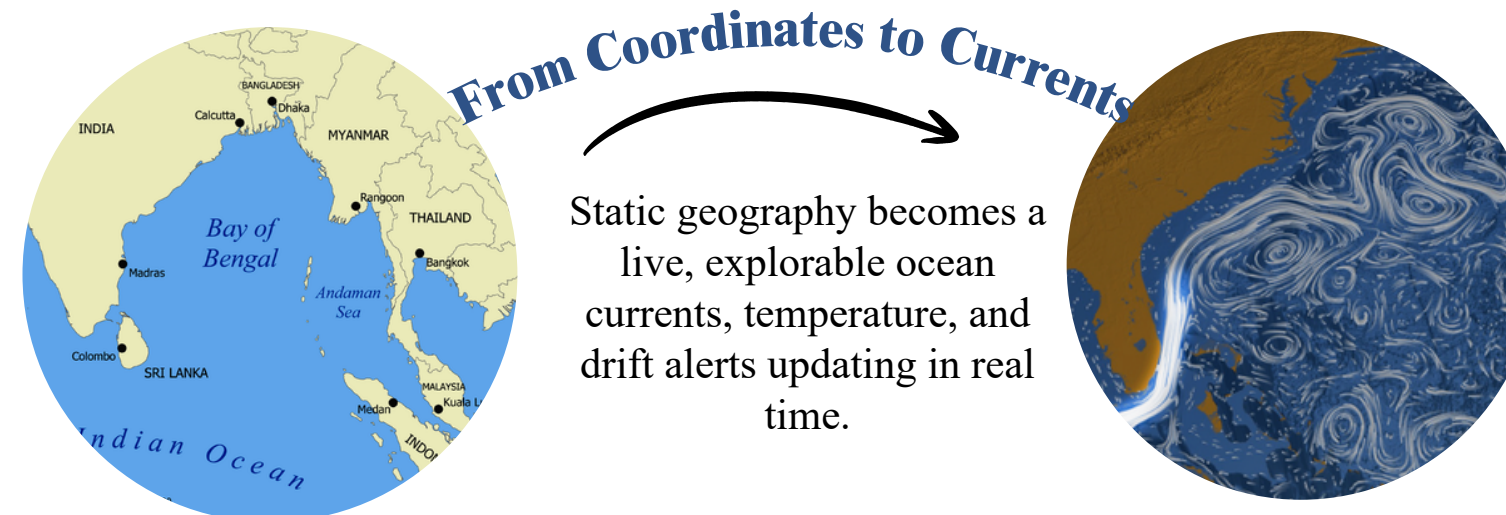
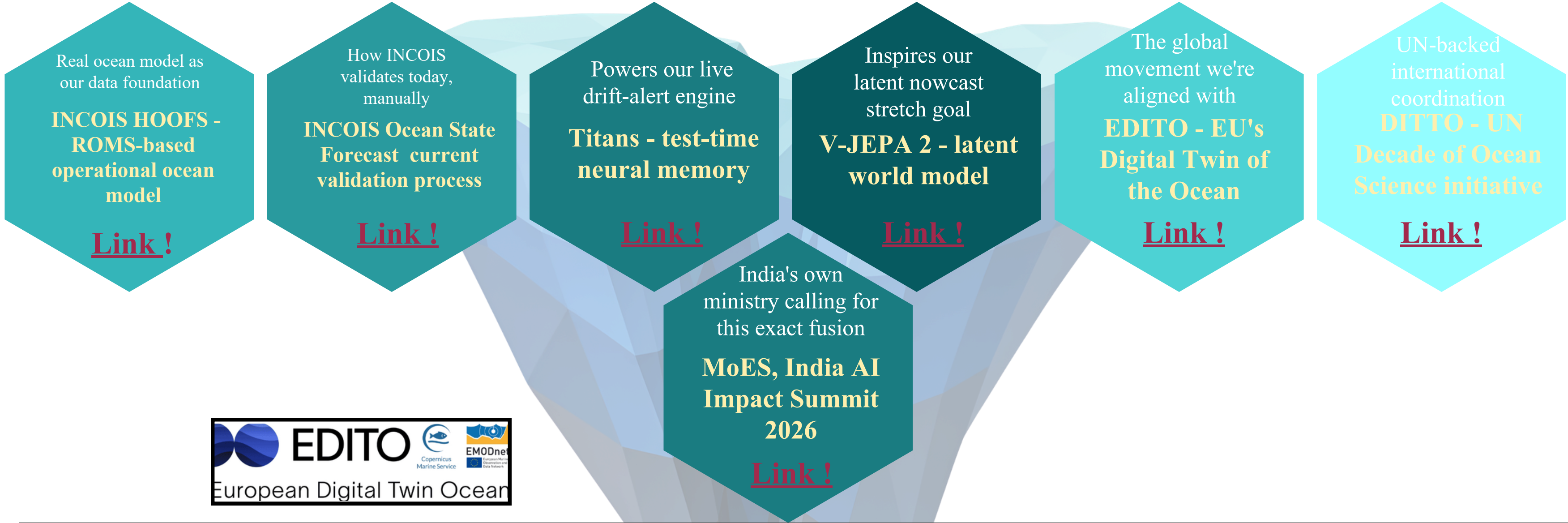


fig. the global Digital Twin Ocean concept



RESEARCH AND REFERENCES



| Dimension                                                    | Existing solutions (INCOIS OSF,ODV,Panoply) | VARUNA               |
|--------------------------------------------------------------|---------------------------------------------|----------------------|
| Web-based 3D, depth-resolved views?                          | No / Partial                                | Yes                  |
| Unified Argo/Glider/CTD/BGC + model, one view?               | No                                          | Yes                  |
| Interactive controls (var depth-slice, time-step, colorbar)? | No                                          | Yes                  |
| Add new sensor/variable without re-engineering?              | No                                          | Yes - adapter plugin |
| Expert + public dual-mode?                                   | No                                          | Yes                  |
| Supports fast operational decision-making?                   | Slow, Manual                                | Demo-ready           |